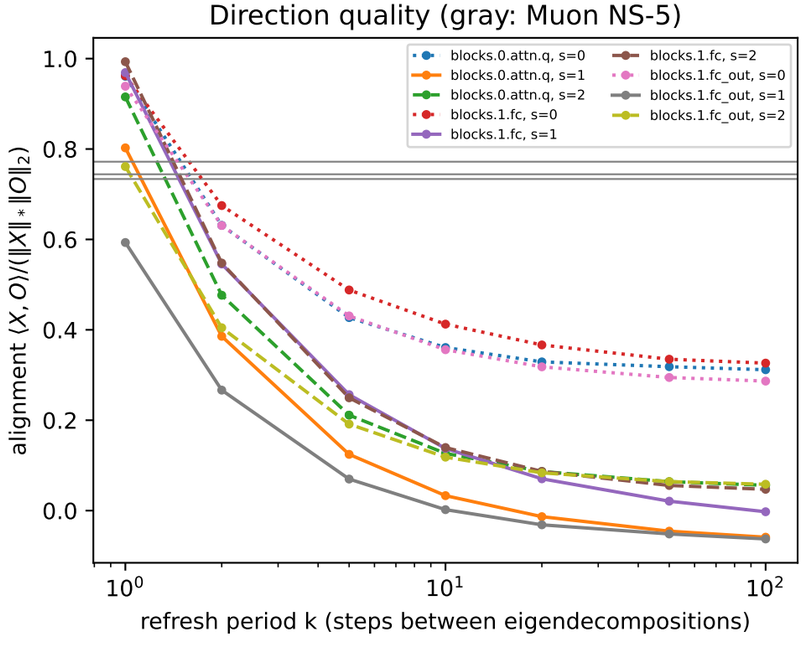
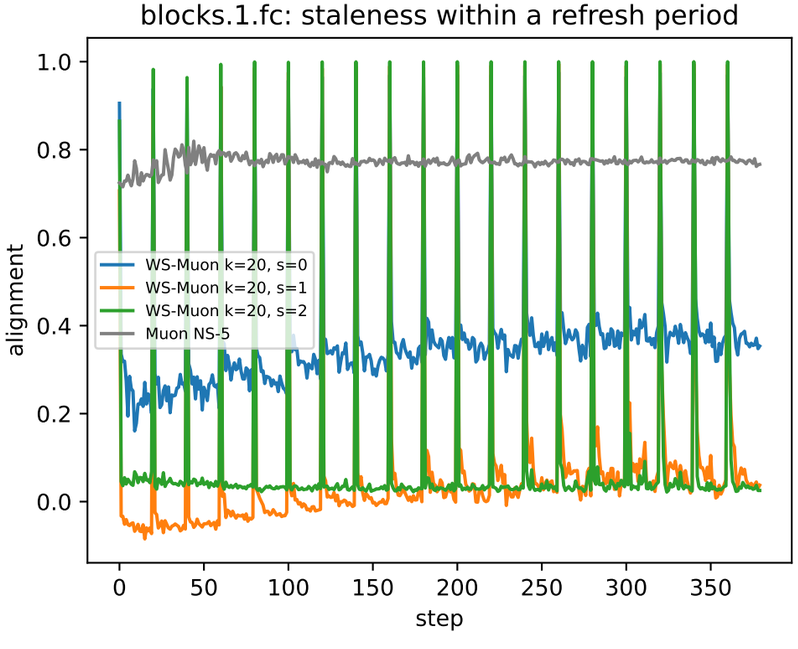
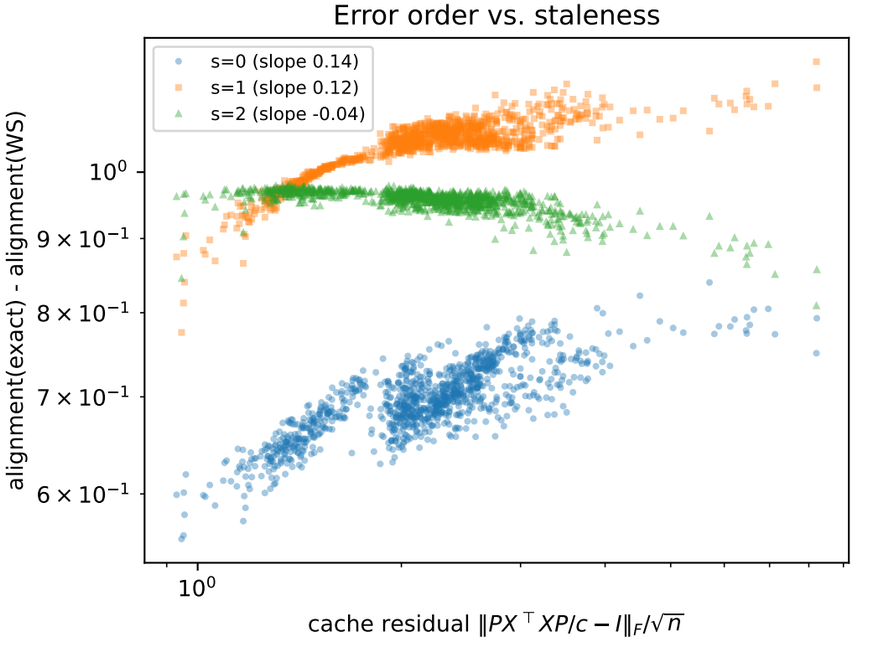

Direction quality (gray: Muon NS-5)
blocks.0.attn.q, s=0
blocks.0.attn.q, s=1
blocks.0.attn.q, s=2
blocks.1.fc, s=0
blocks.1.fc, s=1
blocks.1.fc, s=2
blocks.1.fc_out, s=0
blocks.1.fc_out, s=1
blocks.1.fc_out, s=2
alignment $\langle X, O\rangle/(\|X\|_* \|O\|_2)$
refresh period k (steps between eigendecompositions)


blocks.1.fc: staleness within a refresh period
WS-Muon k=20, s=0
WS-Muon k=20, s=1
WS-Muon k=20, s=2
Muon NS-5
alignment
step


Error order vs. staleness
s=0 (slope 0.14)
s=1 (slope 0.12)
s=2 (slope -0.04)
alignment(exact) - alignment(WS)
cache residual $\|PX^\top XP/c - I\|_F/\sqrt{n}$